

P SURYA SAMYOG

NIT Calicut, Kerala, India

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EDUCATION

National Institute of Technology Calicut <i>Bachelor of Technology in Materials Science and Engineering - CGPA: 8.64</i>	2021 – 2025 <i>Kozhikode, Kerala</i>
Central Board of Secondary Education (CBSE) <i>10th Grade - Percentage: 97.6, 12th Grade - Percentage: 95.2</i>	2018 – 2021 <i>Alappuzha, Kerala</i>

POSTERS AND PRESENTATIONS

Lifelong Machine Learning Potentials for 2D-TMD Aqueous Systems Using EGNNs <i>P Surya Samyog, School of Energy and Environment, City University</i>	July 2025 <i>Hong Kong</i>
DFT-driven Generation of ANN Potentials for 316LN Stainless Steel <i>P Surya Samyog, International Conference on Materials for Energy, Environment & Healthcare</i>	Dec 2024 <i>NIT Calicut</i>

RESEARCH EXPERIENCE

VISTA Lab, RBCCPS, Indian Institute of Science ¹ <i>Project Assistant – Supervisor: Prof. Punit Rathore</i>	Jan 2026 – Present <i>Bengaluru</i>
VISTA Lab, RBCCPS, Indian Institute of Science ¹ <i>Research Intern – Supervisor: Prof. Punit Rathore</i>	Nov – Dec 2025 <i>Bengaluru</i>
School of Energy and Environment, City University <i>Visiting Research Fellow – Supervisor: Prof. Patrick Sit</i>	July 2025 <i>Hong Kong</i>
Computational Nanotechnology Laboratory (CNL), MSED ² <i>Undergraduate Researcher – Supervisor: Prof. Jeetu S. Babu</i>	Aug 2024 – May 2025 <i>NIT Calicut</i>
Applied Nano Engineering Laboratory, MSED <i>Undergraduate Researcher – Supervisor: Prof. V. Sajith</i>	Jan – May 2024 <i>NIT Calicut</i>

RESEARCH PROJECTS

Self-supervised Learning on Graphs using Sheaf Theory ¹ <i>VISTA Lab, IISc Bengaluru</i> - Developing a novel sheaf-based graph self-supervised learning framework, aiming for ICML and DMLR submissions. - Conducting baseline experiments with existing graph SSL models to assess the embedding quality for downstream tasks. - Working toward building a theory for a sheaf-based graph SSL method and also achieving better computational efficiency on larger graphs.	Nov 2025
First-Principles Investigation of VOC Adsorption on Ti3C2Tx MXenes ² <i>CNL, NIT Calicut (Remote)</i> - Modelled MXene structures using Materials Studio. - Conducting theoretical experiments using Quantum Espresso to support the experimental findings.	Sep 2025
Equivariant Graph Neural Networks with Higher-Order Node Interactions <i>Independent Project</i> - Implemented an EGNN model from scratch using PyTorch and PyTorch Geometric. - Developed theory for a framework with three-body interactions between atoms (nodes) and implemented it using PyTorch. - Conducted experiments on QM9 dataset with the new framework which outperformed the earlier one in preserving structural symmetry.	May 2025
Enhancement of HER performance of 2D ReS₂ via Bimetallic Nanoclusters: A Machine Learning-Assisted DFT Study ² <i>CNL, NIT Calicut</i> - Investigated the electrocatalytic properties of pure rhenium sulfide monolayer using DFT simulations on Quantum Espresso. - Formulated theory explaining the catalytic inactivity of the monolayer with a series of property calculations. - Developing a database of bimetallic transition metal nanoclusters (13 atoms, Icosahedral geometry) with identified adsorption sites and optimised structures to train a predictive ML model for high-throughput screening of metal nanoclusters.	Jan 2025
Physics-Informed Neural Networks for Solving Nonlinear PDEs using DeepXde <i>NIT Calicut</i>	March 2025

- Implemented inverse PINNs to solve the Navier-Stokes equations by identifying unknown flow parameters from time-resolved 2D velocity and pressure field data of fluid flow past a cylinder. - Modelled and visualised phase separation dynamics using the Allen-Cahn equation. - Simulated single soliton wave propagation by solving the Korteweg–de Vries (KdV) equation using PINNs.	
DFT-Driven Generation of ANN Potentials for 316LN SS using ænet ²	Aug 2024
<i>CNL, NIT Calicut</i>	
- Developed a Python-based workflow to automate DFT dataset generation using Atomic Simulation Environment (ASE). - The ænet package was used to develop an ML model to generate ANN potentials.	

FELLOWSHIPS AND GRANTS

Summer Research Fellowship, City University of Hong Kong	July 2025
<i>Hosted by School of Energy and Environment</i>	<i>Hong Kong</i>
One of the 9 selected students from India for the research programme.	
Student Innovative Project Grant	2024-25
<i>Granted by Centre for Innovation, Entrepreneurship and Incubation (CIEI)</i>	<i>NIT Calicut</i>
- Project: Application of Hyperspectral Imaging and Machine Learning for Quality Assessment of Pepper - Role: Team member	

INDUSTRIAL EXPERIENCE

ZFWPC – ZF Friedrichshafen AG	May – July 2024
<i>Data Analysis and Process Engineering Intern</i>	<i>Coimbatore, Tamil Nadu</i>
- Conducted research on lean management and production process flow and constructed a current state VS mapping to identify inefficiencies, reduce lead time, and minimise waste. - Developed DAX measures to analyse large datasets, enabling real-time inventory tracking.	

CONFERENCES AND RESEARCH VISITS

MTC Srolovitz Group, The University of Hong Kong	July 2025
<i>Department of Mechanical Engineering</i>	<i>Hong Kong</i>
MEEHCON’24	December 2024
<i>International Conference on Materials for Energy, Environment & Healthcare</i>	<i>NIT Calicut</i>

CERTIFICATIONS AND ADDITIONAL COURSES

IBM AI Engineering Professional Certificate	Present
<i>Coursera</i>	<i>Online</i>
Basic Numerical Methods	Present
<i>TU Dresden</i>	<i>Online</i>
DFT Modelling of Advanced Materials (Hands-on-Training)	March 2025
<i>Centre for Advanced Computational Research</i>	<i>New Delhi</i>

LEADERSHIP / EXTRACURRICULAR

The Literary and Debating Club	Oct 2023 – Oct 2024
<i>General Secretary</i>	<i>NIT Calicut</i>
Led initiatives on mental health and Pride, revived <i>Kaapikootam</i> open house discussions, organised cultural and literary events, laid the groundwork for Sahiti literature fest, and fostered a collaborative club environment.	
Kerala Literature Fest 2024	January 2024
<i>Volunteer</i>	<i>Kozhikode</i>
Volunteered at Asia’s largest literature festival, leading a team of volunteers and ensuring smooth guest–volunteer communication while addressing guest needs.	
Tathva 2023	Aug 2023 – Jan 2024
<i>Design Team Head</i>	<i>NIT Calicut</i>
Led a team of graphic designers, creating social media posters for the fest and fostering a collaborative environment to ensure timely delivery.	
Miraqui – Fashion Show Team	April 2023 – Oct 2024
<i>Captain and Marketing Head</i>	<i>NIT Calicut</i>
Coordinated a large team to ensure smooth collaboration, organised model practice sessions, and managed team finances, including preparing the annual budget.	

CREATIVE WORK

The Literary and Debating Club	Sep 2022 – Oct 2024
<i>Graphic Designer & Illustrator</i>	<i>NIT Calicut</i>
The Brand Project - Freelance	June 2023
<i>Brand Designer</i>	<i>Remote</i>